

SERVICE MANUAL

Invivo Corporation
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Document 2380635-4

Revision 2

GE Signa[®] EXCITE[™] 3.0T Quadrature Extremity Coil

GE Catalog Part Number: M1385AM

Supplier Part Number: 14280

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Rev. October 28, 2009

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(JA)

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(KO)

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- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
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(LV)

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- Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdīnācija neievērošanas rezultātā var rasties elektriskās strāvas triecienu, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

(LT)

Šis eksploatavimo vadovas yra tik anglų kalba.

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- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

(NO)

Denne servicehåndboken finnes bare på engelsk.

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(PL)

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- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanimarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

TABLE OF CONTENTS

Section 1 - Introduction.....	1
1-1 Product Identification and Shipping List	1
1-2 Compatibility	1
1-3 Related Documentation	1
1-4 Environmental Requirements.....	2
1-5 Theory of Operation.....	2
Section 2 - Setup and Calibration	4
2-1 Coil Installation	4
2-1-1 Special Install Notes	4
2-1-2 Installing the Coil	4
2-2 Installation Functional Checks	4
2-3 Periodic Quality Assurance Check.....	4
Section 3 - Functional Checks.....	5
3-1 Scanner Verification.....	5
3-2 Coil Imaging Performance Verification.....	5
3-2-1 Tools Required	5
3-2-2 Explanation of Procedure	5
3-2-3 Signal Scan	5
3-2-4 Noise Scan	7
3-2-5 SNR Image Analysis.....	9
3-3 External Cable Check	10
3-4 PIN Diodes Check	13
3-5 Mechanical Hardware Check	13
3-6 Troubleshooting Tips	14
3-6-1 Receiving No Signal	14
3-6-2 Image Quality	15
3-6-3 Artifacts	16
Section 4 - Maintenance.....	17
4-1 Coil Care.....	17
4-2 Special Care Requirements.....	17
Section 5 - Replacement	18
5-1 Disassembly of Coil	18
5-2 External Cable Replacement	18
5-3 Mechanical Hardware Replacement	19
Section 6 - Renewal Parts.....	20
6-1 Field Replaceable Units	20
6-2 Other Replaceable Accessories.....	22
Section 7 - Appendix	23
7-1 SNR Data Sheet	23
7-2 Schematic.....	24
7-3 Coil Configuration	25
Revision History	26

SECTION 1 - INTRODUCTION

1-1 Product Identification and Shipping List



NOTE: Coils do not ship with phantoms. Phantoms come in a unified phantom set with the MR system

SHIPPING LIST – TABLE 1-1

Description	GE Part#	Supplier Part#	Qty
GE Signa EXCITE 3.0T Quadrature Extremity Coil	2380635-2	14280	1
Knee Pad	2357819-3	14629	1
Leg/Foot Support Pad	2357819-4	14627	1
Foot Support Pad	2357819-5	14626	1
Toe Pad	2357819-6	14628	1
Phantom Positioner (part numbers are equivalent)	5147225-7 / 2357819-11	14270	1
Operator's Manual	2380635-3	2380635-3	1
Service Manual	2380635-4	2380635-4	1

1-2 Compatibility

This coil is compatible with the following hardware configuration:

- GE Signa EXCITE 3.0T G3 Systems
- GE Signa EXCITE 3.0T HD Systems
- GE Signa 3.0T HDx/HDxt Systems (upgrade configuration only)

1-3 Related Documentation

- Signa EXCITE 3.0T 11.x Service Methods - 2389160-1
- Signa EXCITE HD 3.0T 12.x Service Methods - 5133330-1
- Signa HDx 1.5T and 3.0T Service Methods - 5155500

- Signa HDxt 1.5T and 3.0T Service Methods - 5440000
- GE Medical Systems Signa EXCITE 3.0T Quadrature Extremity Coil Operator Manual, 2380635-3

1-4 Environmental Requirements

Storage Requirements

Store the coil in an air-conditioned scan room or equipment room.

To store the coil, a storage space of (55cm) x (45cm) x (40cm) is required.

Dimensions

19" Wide X 12" Deep X 12" High (48.26 cm Wide X 30.48 cm Deep X 35.56 cm High)

Weight

15 lbs (6.81 Kg)

1-5 Theory of Operation

The Quadrature Lower Extremity Coil is a twelve (12) rod transmit-receive birdcage design. In MR systems, a birdcage coil is a near field antenna, which generates a homogeneous transmit RF magnetic field within its volume and, based on the reciprocity principle, has the same homogeneity as an RF receiver. The transmit field can be either linearly or circularly polarized, and is for this device, circularly polarized.

The coil consists of 2 distinct parts: The system independent antenna (which is comprised of the Bottom Coil and the Top Coil) and the cable section (which contains the Combiner and Bazooka Balun). Refer to Appendix 7-2.

During the transmit phase, RF power is provided to the transmit port (TX) of the system connector and the power is passed to the hybrid combiner. The power is split into two signals of equal amplitude and 90 degree phase difference (Mode 1 and Mode 2). These signals pass through the Bazooka Balun to remove any standing waves. The signals are then coupled to the antenna via a match and tune circuit and a balun circuit and generate a circularly polarized magnetic field useable for MRI. During the receive or detection phase, the MRI signal is received by the antenna and coupled to the system cable via the balun circuit and the match and tune circuit passing through the Bazooka Balun to the Combiner and then routed to the receive port (RX) of the system connector.

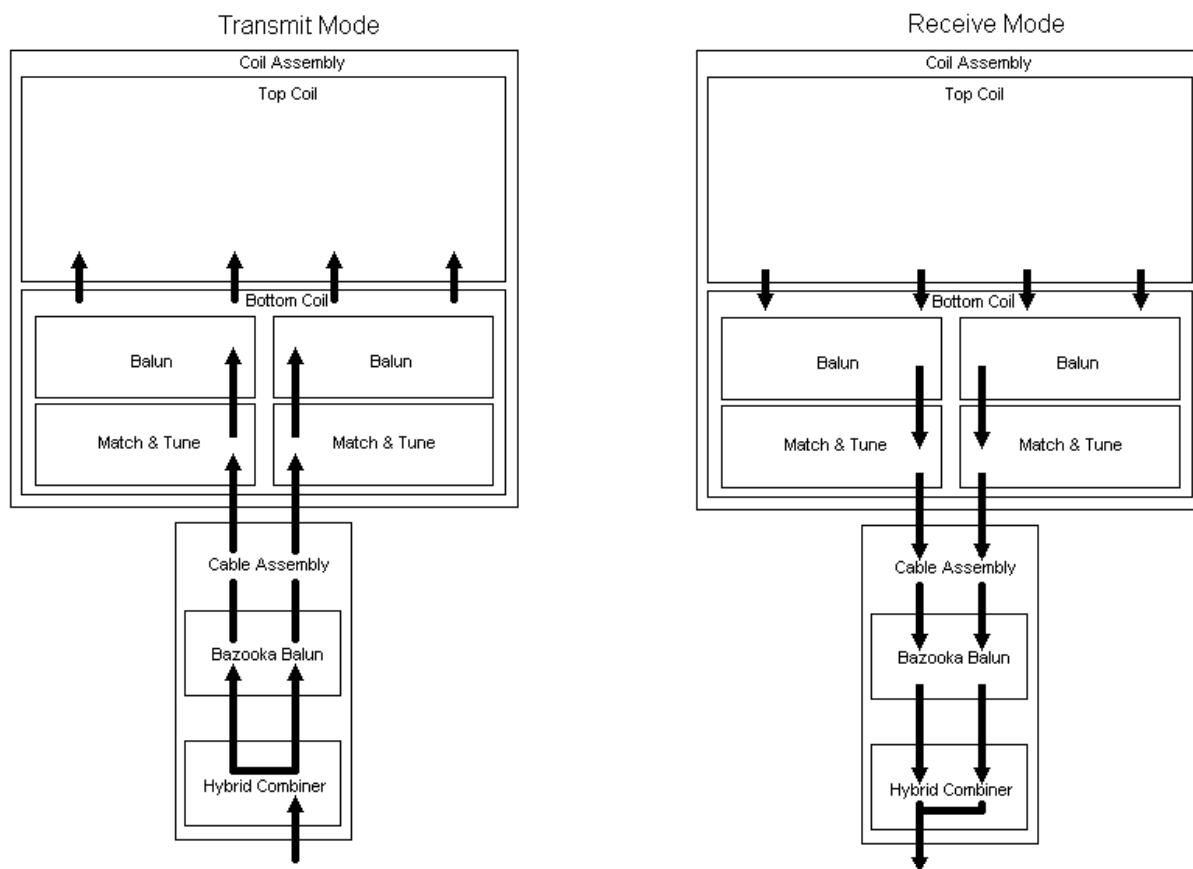


Figure 1-5: Signa EXCITE 3.0T Extremity Coil Block Diagram

SECTION 2 - SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

Check that Coil Polarity matches System Polarity. Coils are shipped for forward ramped magnets. For reverse ramped magnets, see comments in Section 5-2.

2-1-2 Installing the Coil

The name for this coil is: **Quadknee**

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; Software; System Level Software Utilities; Configuration File Manager.

If the coil does not exist in the Coil Config File refer to the Adding New Coils to Config File Manager procedure and use the coil configuration information in Section 7-3 of this manual.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; Functional Checks; System Level Functional Checks; Signal to Noise Check.
2. Perform Section 3-2 - Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coils is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 - Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 - FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

Note that the SNR test can be performed using either a legacy phantom set that had shipped with the coil or a unified phantom set which ships with the MR system.

3-2-1 Tools Required

TOOLS REQUIRED USING LEGACY PHANTOM/POSITIONER – TABLE 3-2-1

Description	GE Part#	Supplier Part#	Qty
Signa EXCITE 3.0T Extremity Coil Phantom	2357819-12	14246	1
Signa EXCITE 3.0T Extremity Coil Phantom Positioner	2357819-11	14270	1

TOOLS REQUIRED USING UNIFIED PHANTOM/POSITIONER – TABLE 3-2-1

Description	GE Part#	Supplier Part#	Qty
Large cylindrical Unified Phantom, SiOil	5342679-2	N/a	1
Phantom positioner (legacy and unified are the same)	5147225-7 / 2357819-11	N/a	1

3-2-2 Explanation of Procedure

The Signa EXCITE 3.0T extremity coil can be used in one (1) mode of operation and has one (1) coil name: **QUADKNEE**. Refer to the Data Sheet in Appendix 7-1 to understand the data required to calculate the individual element SNR.

The image quality check uses two different protocols for signal and noise image acquisition. The signal scan is an **FSE** sequence used to minimize susceptibility and B₀ inhomogeneity effects. The noise scan is a **GRE** sequence that has a Control Variable (do_noise) to eliminate the transmit RF completely during the scan. The signal scan **must** be run prior to the noise scan as the R1, R2, and TG values from the signal scan are used for the noise scan.

3-2-3 Signal Scan

1. From the Scan Desktop, start new scan by selecting **[New Pt]**; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click **[Patient Position]** to open protocols window.
2. Remove all other surface coils from the cradle. Place the coil on the cradle as illustrated in Figure 3-2-3-1. The coil should be positioned such that its cables exit towards the magnet. Remove any pads from the coil. Set the phantom positioner into the coil as seen in Figure 3-2-3-1.



Figure 3-2-3-1: Placement of posterior section and phantom positioner

3. Center the coil within the baseplate and place the phantom in the phantom positioner as shown in Figure 3-2-3-2.

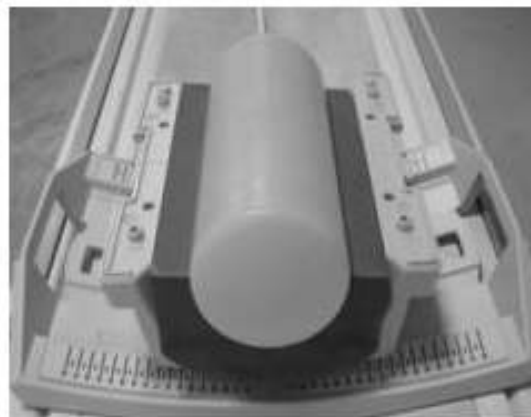


Figure 3-2-3-2: Phantom position. Left image is legacy phantom and right is unified phantom.

4. Attach the anterior portion of the coil to the baseplate and lock the coil as shown in Figure 3-2-3-3.

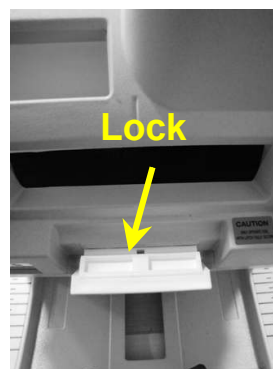


Figure 3-2-3-3: Anterior portion connected (shown in unified phantom).

5. Connect the Coil to the respective Port of the system LPCA and move the cradle to align the coil and landmark on the marks shown in Figure 3-2-3-4.



Figure 3-2-3-4: Coil alignment and landmarking

6. Move the coil to scan position by pushing the “**Move to Scan**” button, ensuring cable does not get snagged.
7. At the console, set the protocols per the Signal section from Table 3-2-4: Signal and Noise Protocols. GE Service Personnel can use the Extremity SNR Signal protocol in the GE, Other protocols section.
8. Click [**Save Series**] to download the protocols, then click [**Download**].
9. Run [**Auto Prescan**]. Take note of the R1, R2 and TG values.
10. Run [**Scan**]

3-2-4 Noise Scan

A signal scan must be run **prior** to the noise scan as the same R1, R2 and TG values must be used for both the signal and noise scans. Do **not** run an Auto Prescan prior to the noise scan as the values will be changed.

1. Copy the signal scan series. Use [**Copy Series**] (highlight signal series and click right mouse button) and [**Paste Series**] in **RX Manager**.
2. Click [**View Edit**] and set the protocols per the Noise section from Table 3-2-4: Signal and Noise Protocols. GE Service Personnel can use the Extremity Coil SNR Noise protocol in the GE, Other protocols section.
3. Click [**Save Series**] and click [**Download**].
4. Open [**Display CVs**] menu under [**Research Operations**]. Set the “rhformat” and “do_noise” CVs to “1”.
5. Run [**Manual Prescan**], and do **not** make any changes. Verify that R1, R2, TG and Bandwidth have not changed from the previously completed signal scan, and click [**Done**].
6. Run [**Scan**].

SIGNAL AND NOISE PROTOCOLS – TABLE 3-2-4

Protocol	Signal	Noise
Protocol	Signal	Noise
<i>Patient Exam Information</i>		
Patient ID	geservice	geservice
Patient Name		
Patient Weight	111 lbs. (50 kg)	111 lbs. (50 kg)
<i>Patient Position</i>		
Patient Position	Supine	Supine
Patient Entry	Head First	Head First
Coil	QUADKNEE	QUADKNEE
Series Description	Signal	Noise
<i>Imaging Parameters</i>		
Plane	Axial	Axial
Mode	2D	2D
Pulse Seq	FSE-XL	GRE
Imaging Options	Fast (or None)	None
PSD Name	<i>Leave Blank</i>	<i>Leave Blank</i>
Protocol	<i>Leave Blank</i>	<i>Leave Blank</i>
<i>Scan Timing</i>		
# of Echoes	1	1
TE	17.0	Min Full
TR	500.0	34
Echo Train Length	4	N/A
Bandwidth	15.63	15.63
<i>Additional Parameters</i>		
Flip Angle	N/A	1
<i>Acquisition Timing</i>		
Freq	256	256
Phase	256	256
NEX	1.00	1.00
Phase FOV	1.00	1.00
Freq DIR	R/L	R/L
Auto Center Freq	Peak	Peak
Autoshim	On	On
Phase Correct	On	N/A
Contrast	Off	Off
# of Reps B4 Pause	0	N/A
<i>Scanning Range</i>		
FOV	20	20
Slice Thickness	3.0	3.0
Spacing	1.5	1.5
S/I Start	0.0	0.0
R/L Center	0.0	0.0
A/P Center	0.0	0.0
S/I End	0.0	0.0
# Slices	1	1
Table Delta	0.00	0.00

3-2-5 SNR Image Analysis

SNR Measurement

A circular ROI should be used for the signal measurements, and a rectangular ROI should be used for the noise measurements. The actual ROIs should be similar in shape and size to those in Figure 3-2-5 below.

ROIs in both signal and noise images can be measured directly in the image browser. Click the user interface button [MEASURE], select the circular or rectangular shape, and adjust its size and orientation so that the ROI is similar to those found in Figure 3-2-5. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image.

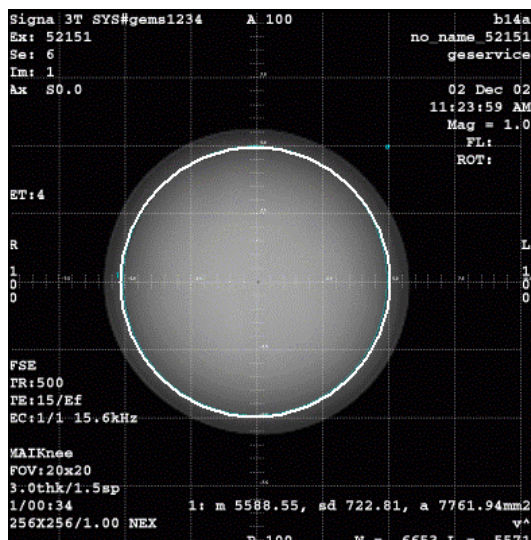
For the signal measurement, choose a 10 cm diameter circular ROI. This should be approximately a 5600 mm². For the noise measurement, choose a 19 cm rectangular ROI. This should be approximately 34000 mm².

Record the values for mean signal, standard deviation of noise, and SNR in Table 7-1.

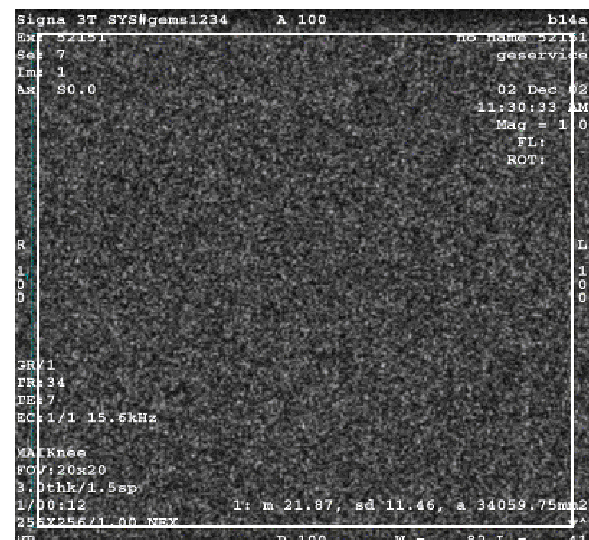
Individual receiver SNR is defined as the mean of data within the signal ROI divided by the standard deviation of data within the noise ROI.

$$SNR = \frac{\text{Mean of Signal within ROI}}{\text{Std.Deviation of Noise within ROI}}$$

Note: The SNR calculation uses the **MEAN** of the signal image and **STANDARD DEVIATION** of the noise image.



A: Signal ROI Measurement (10cm dia.)



B: Noise ROI Measurement (19cm)

Figure 3-2-5: Example ROIs

SNR Specification

The SNR measurements must be greater than or equal to the following specifications:

SNR SPECIFICATIONS – TABLE 3-2-5

Plane (if applicable)	SNR
Signa EXCITE 3.0T Quad Extrem Coil	410 (Legacy Phantoms) 207 (Unified Phantoms)

3-3 External Cable Check

1. Remove four (4) screws from the Bottom Cover of the Interface Box Assembly.
2. Remove the Top Coil Assembly.
3. Remove four (4) large nylon screws from the Bottom Coil Assembly.
4. Carefully lift the Bottom Inner Assembly out of the Bottom Cover and set it down on the Cable Assembly.
5. Using the DMM, verify continuity of the cable by performing the following operations:

Note: Flex the cable while checking to test for intermittent opens or shorts in the cable.

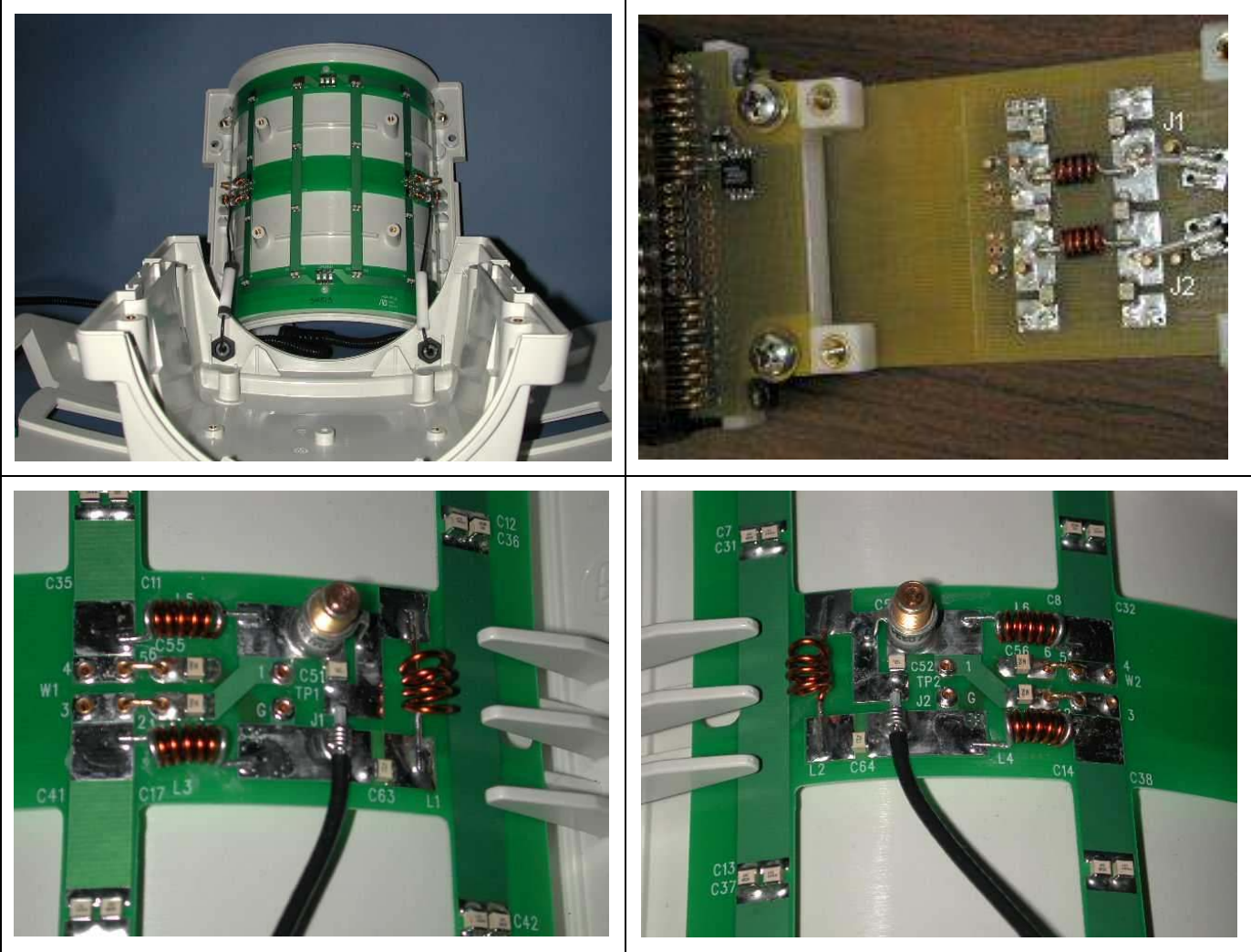
- a) A continuity check for Receive should be measured from J2 center pin on the Interface Box PCB to the J1 center pin on the Bottom Coil PCB.
 - b) A continuity check for Receive should be measured from J2 Gnd on the Interface Box PCB to the J1 Gnd on the Bottom Coil PCB.
 - c) A continuity check for Transmit should be measured from J1 center pin on the Interface Box PCB to the J2 center pin on the Bottom Coil PCB.
 - d) A continuity check for Transmit should be measured from J1 Gnd on the Interface Box PCB to the J2 Gnd on the Bottom Coil PCB.
 - e) On the Interface Box PCB, verify open (No short circuit) using DMM between J1 center pin and J1 Gnd.
 - f) On the Interface Box PCB, verify open (No short circuit) using DMM between J2 center pin and J2 Gnd.
6. If the cable fails either test, remove the cable assembly per Section 5-2, External Cable Replacement. Affix the Bottom Cover of the Interface Box Assembly using four (4) screws and return for repair.
 7. If the cable passes the tests, affix the Bottom Cover of the Interface Box Assembly using four (4) screws.
 8. Carefully fit the Bottom Inner Assembly into the Bottom Cover.

9. Install four (4) large nylon screws from the Bottom Coil Assembly.

10. Seat the Top Coil Assembly to the Bottom Coil Assembly.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3

Signal	Positive Lead	Negative Lead	Reading
	J2 center pin on the Interface Box PCB	J1 center pin on the Bottom Coil PCB	0 Ohms
	J2 Gnd on the Interface Box PCB	J1 Gnd on the Bottom Coil PCB	0 Ohms
	J1 center pin on the Interface Box PCB	J2 center pin on the Bottom Coil PCB	0 Ohms
	J1 Gnd on the Interface Box PCB	J2 Gnd on the Bottom Coil PCB	0 Ohms
	J1 center pin on the Interface Box PCB	J1 Gnd on the Interface Box PCB	Open
	J2 center pin on the Interface Box PCB	J2 Gnd on the Interface Box PCB	Open



3-4 PIN Diodes Check

N/A

3-5 Mechanical Hardware Check

Lift the coil locks on both sides of the coil, and slide the coil from side to side. The coil must slide freely back and forth on the base with no binding or grinding. Lock the coil in the center of the base. Check the bottom of the base to ensure the 4 rubber feet are tight.

3-6 Troubleshooting Tips

The following is a list of common problems and solutions for those problems.

3-6-1 Receiving No Signal

Problem:

You are unable to pre-scan **or** are scanning and yet receiving no signal.

Possible Solution:

Verify that you have selected the appropriate system coil selection. Refer to Section 2-1-2 for additional information.

Verify the cable is correctly connected to the system and that it is completely plugged in.

Verify that the landmark is correct and that the cradle has not unlatched.

Verify that the scan locations and any FOV offsets are correct.

Perform a Continuity Check on the external cable (to be performed by a G.E. authorized Service Engineer only).

If you still cannot get a signal, try to scan (transmit and receive) with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still receive no signal the problem probably lies with your MR system. If the body coil scan is satisfactory, attempt a scan using both another coil of the exact same type (receive-only, phased array or transmit/receive, whichever applies) and the same system coil selection. If the scan completes successfully, there is probably a problem with the Invivo coil. Contact Invivo Corporation for further assistance. If you are unable to scan with the substitute coil, there may be a system problem related to this particular coil type.

3-6-2 Image Quality

Problem:

The ratio obtained in the periodic quality assurance check is not greater than 85%, or the image quality is not what you expect it should be, given the parameters selected.

Possible Solution:

Review the selected protocol. If you are performing the Periodic Quality Assurance, be sure your protocol is **identical** to the protocol provided on Table 3-2-4 of this manual. If you are performing diagnostic images, you may need to increase NEX or FOV.

Perform a Continuity Check on the external cable (to be performed by a G.E. authorized Service Engineer only).

Verify that there are no loops in the cables.

Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).

Verify that the coil is properly positioned.

Verify that your center frequency is within the frequency adjustment range for your system.

Verify that the R1, R2 and TG values from the pre-scan are within normally expected ranges.

Cleaning the electrical contacts that connect the upper and lower halves with Isopropyl Alcohol and a cotton swab has been demonstrated to improve the coil's SNR on occasion.

If you have not done so already, perform a system Quality Assurance phantom test, as outlined in Section 3 of this manual. If the values you obtain do not fall within normal operating parameters, investigate this further by performing a phantom scan with the body coil. **For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil.** If you still have the same problems, there is probably an MR system problem. If the body coil scan is satisfactory, acquire a scan using **both** another coil of the **exact same type** (receive-only, phased array or transmit/receive, whichever applies) and the same system coil selection. If the image quality is visibly improved, there may be a problem with the Invivo coil. Contact Invivo Corporation for further assistance. If the image quality still suffers, there may be a system problem related to imaging with this type of coil.

3-6-3 Artifacts

Problem:

There is a black line or signal void on the image.

Possible Solution:

Verify that there is no metal present in the area being scanned in or on the patient.

Problem:

Some or all of the images appear shaded or exhibit uneven signal or banding.

Possible Solution:

Confirm that no metallic objects are located nearby, outside the FOV. This is especially important on images utilizing Fat Saturation.

If Fat Saturation is being used, verify that the CFA fine adjustment has been optimized.

For quadrature coils:

Verify that the coil is positioned with the cable exiting towards the bore and that the system and coil polarity match.



Figure 3-2-3-3: Coil alignment

SECTION 4 - MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

The Signa EXCITE 3.0T Quadknee Coil and patient comfort pads **must** be cleaned and stored using the following procedures:

Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.

Do not spray or pour any cleaning solution directly on the coil!

Let the coil housing and pads dry before use.

Store the coil in an air-conditioned scan room or equipment room.

Under no circumstances should the coil be placed into any type of sterilizer!

4-2 Special Care Requirements

N/A

SECTION 5 - REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 Disassembly of Coil

- 1) Open the Draw Latches and remove the Top Coil Assembly and set aside.
- 2) Remove the four large screws that hold the Bottom Inner Coil to the Bottom outer Housing.
- 3) Carefully lift the Bottom Inner Assembly out of the Bottom Cover and set it down on the Cable Assembly.

5-2 External Cable Replacement

Removal

- 1) Perform Section 5.1, Disassembly of Coil.
- 2) Unthread the Connector Cover Caps from the Connector Covers on both cables.
- 3) Separate the connector pins on both cables.
- 4) Carefully set the Bottom Inner to the side.
- 5) Remove the screw in Cable Nut A from both sides.
- 6) While holding the one cable with a 15mm flat, open end wrench, turn Cable Nut B counter clockwise using pliers.
- 7) Repeat for other cable.
- 8) Pull the cable assembly out of the Bottom Housing.

Reinstallation

Note: For Reverse coils the cable with the black Connector Cover Cap enters the mounting hole on the left.

- 1) Insert the new cable into the Bottom Housing ensuring the cable with the black Connector Cover Cap enters the mounting hole on the right.
- 2) Slide Cable Nut A onto the cable and ensure it seats in notch with the threaded hole up, then thread Cable Nut B onto the Cable, using a 15mm flat, open end wrench and pliers. Ensure that when tight, Cable Nut B's notch is centered over the threaded hole in Cable Nut A.
- 3) Repeat for the remaining cable.
- 4) Place the Bottom inner in a position so that the connector pins can be mated together.
- 5) Thread the Connector Cover Cap into the Connector Cover on both cables.
- 6) Carefully set the Bottom Inner into the Bottom Housing and secure with the four large screws that were removed.
- 7) Seat the Top Coil Assembly onto the Bottom Coil Assembly and secure with the Draw Latches.

5-3 Mechanical Hardware Replacement

See Section 6-1, Field Replaceable Units.

SECTION 6 - RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1-1

	Description	GE Part#	Supplier Part#
1	GE Signa® EXCITE™ 3.0T Foot/Knee Assembly w/ Cable	2380635-2	14280
2	Signa® EXCITE™ 3.0T Knee/Foot Cable Assembly	2380635-5	14281
3	Legacy Phantom Assembly	2357819-12	14246
4	Phantom Positioner (two part numbers are equivalent)	2357819-11 or 5147225-7	14270
5	3T Extremity Coil Repair Kit (See Table 6-1-2)	2357819-10	14234
6	Large cylindrical Unified Phantom, SiOil (not shown)	5342679-2	N/a

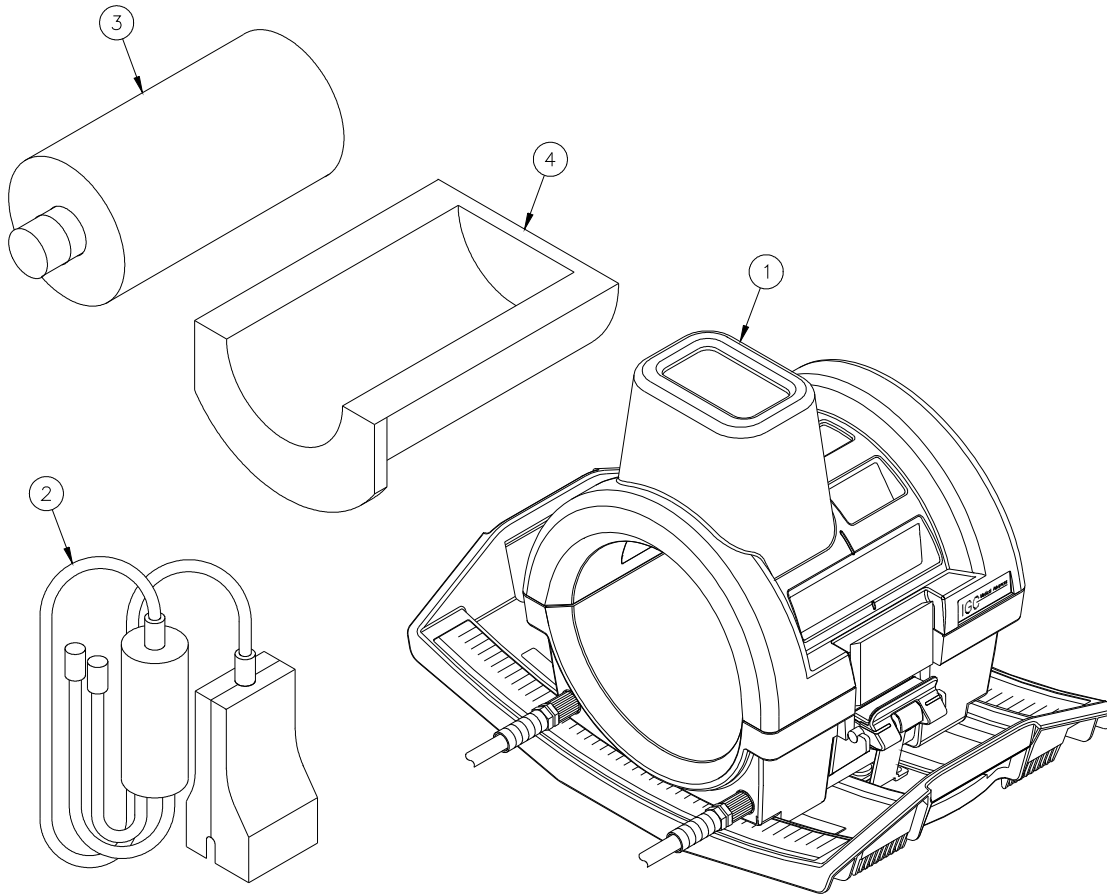


Figure 6-1-1: Field Replaceable Units

CONTENTS OF SPARE PARTS REPAIR KIT – TABLE 6-1-2

Item	QTY	Description	Supplier Part#
1	2	Draw Latch	20074P2
2	4	#6-32 x 0.438 FHMS, Brass	20117P25
3	2	Handle, Coil Lock	19711P1
4	2	Cover, Lock Handle	19712P1
5	4	#4-40 x 0.375 PHMS, Brass	20136P2
6	1	T-Bracket	19713P1
7	3	#10-32 x 0.375 PHMS, Brass	20136P14
8	4	Rubber Bumper	19727
9	4	#6-32 x 0.50 PHMS, Brass	20136P6

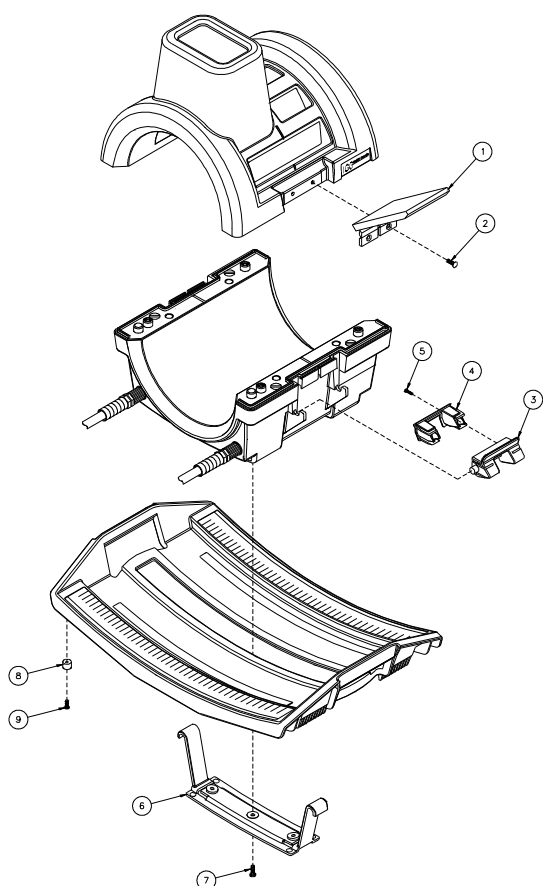


Figure 6-1-2: Spare Parts Repair Kit

6-2 Other Replaceable Accessories

TABLE 6-2: OTHER REPLACEABLE ACCESSORIES LIST

Description	GE Part#	Supplier Part#
Knee Pad	2357819-3	14629
Leg/Foot Support Pad	2357819-4	14627
Foot Support Pad	2357819-5	14626
Toe Pad	2357819-6	14628

SECTION 7 - APPENDIX

7-1 SNR Data Sheet

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

Table 7-1: SNR Data Sheet

[illegible]

$$SNR = \frac{\text{Mean of Signal within ROI}}{\text{Std.Deviation of Noise within ROI}}$$

7-2 Schematic

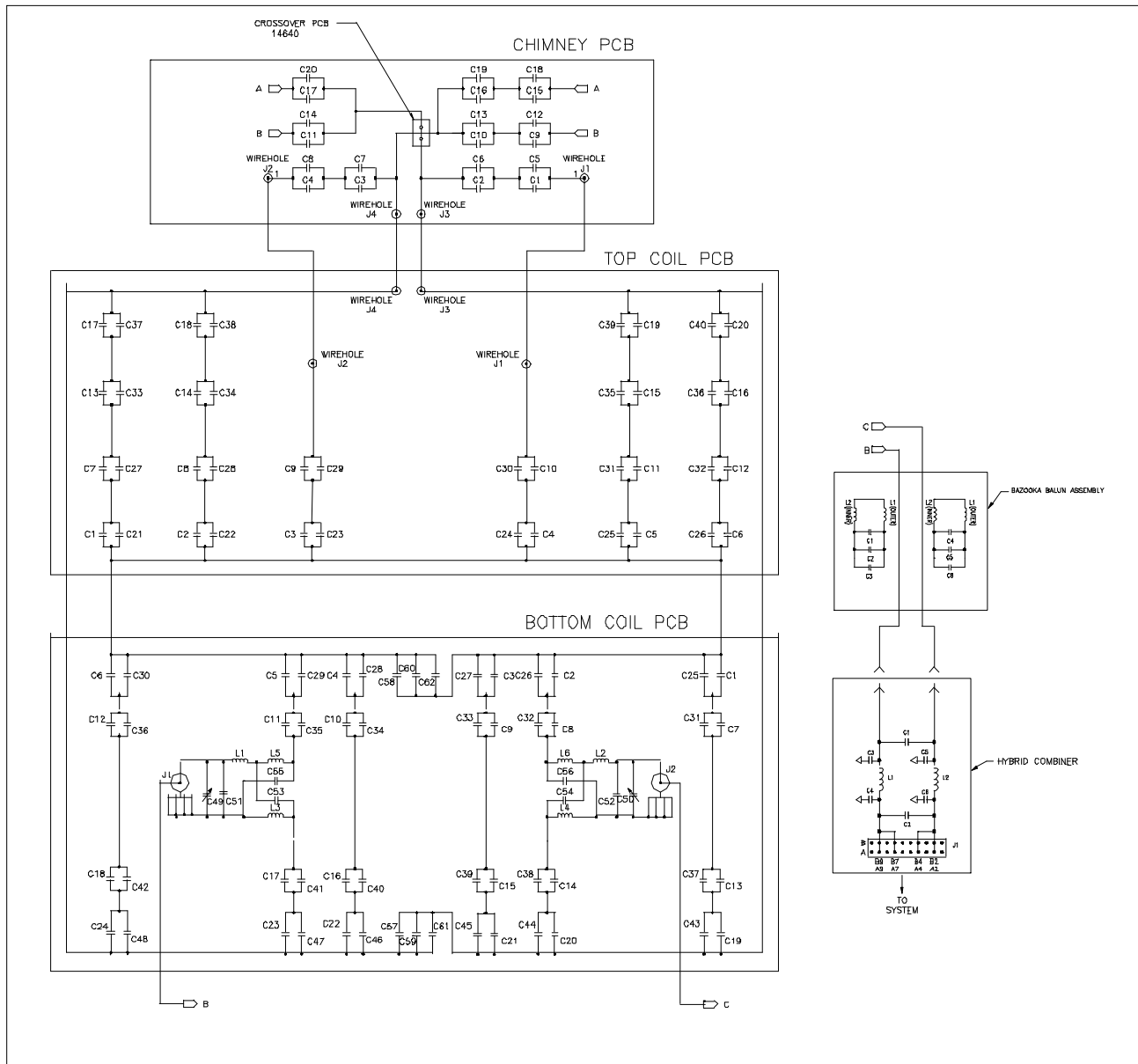


Figure 7-2: Signa EXCITE 3.0T Extremity Coil Schematic

7-3 Coil Configuration

Note: This coil configuration is only current as of the printing of this manual.

Parameter	Signa EXCITE 3.0T
coilName	QUADKNEE
coilCode	ekecd4o92l2ks6mlc7f3d9gkll2aoth6
physicalCoilDescription	Medical Advances T/R knee/foot coil

Parameter	Signa EXCITE 3.0T
coilType	1
multiCoil	no
xmitAtten	27
fastTGstartRG	12
fastTGstartTA	90
cableLoss	1.3
coilLoss	0.064
reconScale	1.00
autosimRcvr	-1
headDefaultFreqDir	1
mCoilSeparationDir	0
maxFOV	480.0
assetThresh	10000
nuclide	1H
scic_a	0
scic_s	0
scic_c	0
mcSwitchSel	0
mcBiasEnable	0
mcVoltConfig0	0
mcVoltConfig1	0
mcOpenCktConfig0	0
mcOpenCktConfig1	0
mcOpenCktConfig2	0
mcOpenCktConfig3	0
mcShortCktConfig0	0
mcShortCktConfig1	0
mcShortCktConfig2	0
mcShortCktConfig3	0

Parameter	Signa EXCITE 3.0T
mcChannelSelect	0
startRec	0
endRec	0
numRec	0
userInterfaceGroup	18
prescanWeight0	1.0
prescanWeight1	1.0
prescanWeight2	1.0
prescanWeight3	1.0
prescanWeight4	1.0
prescanWeight5	1.0
prescanWeight6	1.0
prescanWeight7	1.0
aps1_mod	0
aps1_mod_PStloc	0
aps1_mod_PSrloc	0
aps1_mod_fov	200
aps1_plane	1
opthickPS_mod	5
coilPercentExposed	2
optxpos	5
extremity	yes
mcReconEnable	0
mcPortEnable	0
mcErrorEnable	0
startFastRec	4
numFastRec	1
endFastRec	4
linearQuad	1

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
A	7/3/03	K. Meurer	Engineering pre-release
0	9/26/03	K. Meurer	Original issue. Update coil configuration table.
1	2/18/06	K. Meurer	Update company name.
2	1/6/10	R. David (GE)	Update language policy, shipping list, compatibility, coil imaging performance verification for addition of unified phantom